

Mathematics Stage 3 -- Year B -- Unit 39 Time, Transformations & Angles

KLA: Mathematics | Stage 3 | Year 6 | Term: _____ | Duration: 2-3 weeks | Teacher: _____ | Year: _____

Syllabus outcomes

Code	Outcome descriptor
MAO-WM-01	Develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly.
MA3-GM-03	Measures and constructs angles, and identifies the relationships between angles on a straight line and angles at a point.
MA3-2DS-01	Investigates and classifies two-dimensional shapes, including triangles and quadrilaterals, based on their properties.
MA3-NSM-02	Measures and compares duration, using 12- and 24-hour time.

Syllabus content points (addressed through SOLO tracker)

All content points below are covered by students working independently through the SOLO tracker. Mini masterclasses target specific outcomes based on live tracker data.

Red -- Foundation	Yellow -- On-level	Green -- Extension
Geometric measure A (angles) + foundational angle facts Classify angles as acute, right, obtuse, straight or reflex (GM-A). Measure angles of up to 360° using a protractor (GM-A). Create (draw) angles of up to 360° using a protractor (GM-A). Find missing angles using right angles (90°), a straight line (180°) and a point (360°) (GM-B).	Angle diagrams, transformations & time (applying) Identify right/straight/point angles, vertex and arms, and angle types in diagrams. Use translate, reflect and rotate to describe transformations (position/orientation, not size). Dissect and rearrange one shape to make another. Calculate elapsed time using start/finish times and bridging; round to the nearest minute/hour. Represent commonly used time intervals as decimals. Convert between 12- and 24-hour time and solve duration problems.	Extended -- Stage 4 Angle relationships Use complementary (90°), supplementary (180°) and vertically opposite (equal) angles (MA4-ANG-C-01). Identify corresponding, alternate and co-interior angles on parallel lines cut by a transversal (MA4-ANG-C-01). Solve numerical angle problems using reasoning (MA4-ANG-C-01).

Teaching model

Students work independently through the Word Labs SOLO Tracker (wordlabs.app/solo) across differentiated bands -- Grow (resources), Know (practice), Show (auto-marked assessments). The teacher monitors live dashboard data and pulls small groups for targeted mini masterclasses (15-40 min) based on which outcomes have the most incomplete Show results. Remaining session time is used for incidental one-on-one and small group teaching in response to student questions and tracker data.

Working Mathematically (MAO-WM-01)

Process	How it appears in this unit
Communicating	Students use the language of angles (acute, reflex, vertex, perpendicular) and time (12-/24-hour, duration), and describe transformations.
Understanding and fluency	Students measure and draw angles with a protractor, find missing angles, and calculate elapsed time.
Reasoning	Students reason about angle relationships (90°/180°/360°), that transformations preserve size, and about duration.
Problem solving	Students solve missing-angle and duration problems, and (Stage 4) apply complementary/supplementary, vertically opposite and parallel-line angle relationships.

Differentiation

Support	HPGE / Extension
Students in this group:	Students in this group:

Pre-test

Date	
Format	
Band place ments / key finding s	

Post-test / unit evaluation

Date	
Key finding s	
Studen ts who need further consol idation	
Next steps	

Resources

See Appendix for Resources

Word Labs SOLO Tracker: wordlabs.app/solo | NSW DoE Stage 3 Year B Unit 39 | NESA Mathematics K-10 Syllabus (2022)

Unit 39 -- Outcome Teaching Record

Find the outcome/s you taught, write your notes, sign and date. Order follows the outcome sequence, not your teaching calendar.

Co de	Outcome	Syllabus content point/s addressed	Mini masterclass teaching notes	Sign	Date
R1	Classify angles as acute, right, obtuse, straight or reflex.	GM-A: identify and describe angle size in degrees for the classifications acute, obtuse and reflex MA3-GM-03 Year A	Mini Lesson 1		
R2	Measure angles of up to 360° using a protractor.	GM-A: measure angles of up to 360° using a protractor MA3-GM-03 Year A	Mini Lesson 2		
R3	Create (draw) angles of up to 360° using a protractor.	GM-A: create angles of up to 360° using a protractor MA3-GM-03 Year A	Mini Lesson 3		
R4	Find missing angles on a straight line, at a point and at a right angle.	GM-B: perpendicular lines (90°); adjacent angles add to 90° / 180°; angles at a point add to 360° MA3-GM-03 Year B	Mini Lesson 4		
Y1	Identify angle types and parts (vertex/arms) in diagrams.	GM-B: recognise right/straight/point angles in diagrams; identify the vertex and arms; identify angle types from intersecting lines MA3-GM-03 Year B	Mini Lesson 5		
Y2	Describe transformations (translate, reflect, rotate).	2DS-B: use translate/reflect/rotate; transformations change position and orientation but not size MA3-2DS-01 Year B	Mini Lesson 6		
Y3	Dissect and rearrange one shape to make another.	2DS-B: dissect and rearrange one shape to make another MA3-2DS-01 Year B	Mini Lesson 7		
Y4	Calculate elapsed time using bridging and rounding.	NSM-B: elapsed time from start/finish; add/subtract time using bridging; round to the nearest minute/hour MA3-NSM-02 Year B	Mini Lesson 8		
Y5	Represent commonly used time intervals as decimals.	NSM-B: represent commonly used time intervals as decimals MA3-NSM-02 Year B	Mini Lesson 9		
Y6	Solve duration problems in 12- and 24-hour time.	NSM-B: solve duration problems, including 12-hour and 24-hour notation MA3-NSM-02 Year B	Mini Lesson 10		
G1	Complementary, supplementary and vertically opposite angles.	Stage 4 ANG: complementary (90°), supplementary (180°), vertically opposite angles MA4-ANG-C-01 Stage 4 (Core)	Mini Lesson 11		
G2	Angles on parallel lines cut by a transversal.	Stage 4 ANG: corresponding, alternate and co-interior angles on parallel lines MA4-ANG-C-01 Stage 4 (Core)	Mini Lesson 12		
G3	Solve numerical angle problems using reasoning.	Stage 4 ANG: solve numerical problems involving angles using reasoning MA4-ANG-C-01 Stage 4 (Core)	Mini Lesson 13		

Unit 39 -- Mini Lesson Sequence

Time, Transformations & Angles | Stage 3 Year B

Supplement to the SOLO Teacher Program (pages 1-2). Print pages 1-2 only for annotation. Reference these lesson plans when planning mini masterclasses or when a reviewer requires evidence of planned lesson content.

Red band -- Foundation -- Mini Lessons 1-4

Yellow band -- On-level -- Mini Lessons 5-10

Green band -- Extension -- Mini Lessons 11-13

DoE lesson links: Lessons marked DoE Lesson X draw directly from NSW DoE Mathematics Stage 3 Unit 39. Steps have been condensed to 20-40 minutes for mini masterclass delivery. Resources referenced by their original resource numbers from the DoE unit.

Mini Lesson 1 Classifying angles

R1 | Red band | DoE Lesson 7 (adapted)

Duration: 25 min | **Outcomes:** R1 | Syllabus: MA3-GM-03, MAO-WM-01 | GM-A: classify acute, obtuse, reflex

Linked to: DoE Lesson 7 (adapted) -- Lesson 7 -- Creating, identifying and investigating angles | Key resources: Angle cards, protractors

Learning intention		Success criteria	
Students are learning to: classify angles as acute, right, obtuse, straight or reflex		Students can: * I know an acute angle is less than 90° and a right angle is exactly 90°. * I know an obtuse angle is between 90° and 180° and a straight angle is 180°. * I know a reflex angle is between 180° and 360°.	
Mini masterclass structure (25 min)			
Activate (3 min): Show a right angle. Ask which angles are smaller (acute) and which are bigger (obtuse). Model (10 min): Build the angle types on a number line of degrees: acute <90°, right =90°, obtuse 90-180°, straight =180°, reflex 180-360°. Guided practice (8 min): Students sort angle cards into acute, right, obtuse, straight and reflex. Check (4 min): Exit question: what type of angle is 200°? What about 45°?			
Key vocabulary		Differentiation	
angle, acute, right, obtuse, straight, reflex, degrees		Support: Provide a reference strip showing each angle type with its degree range. Extension: Estimate the size of given angles in degrees before classifying them.	
Assessment		Teaching notes (annotate here)	
Students classify angles — links to R1 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:	

Mini Lesson 2 Measuring angles with a protractor [HANDS-ON]

R2 | Red band | DoE Lesson 7 (adapted)

Duration: 30 min | **Outcomes:** R2 | Syllabus: MA3-GM-03, MAO-WM-01 | GM-A: measure angles up to 360° with a protractor

Linked to: DoE Lesson 7 (adapted) -- Lesson 7 -- Creating, identifying and investigating angles | Key resources: Protractors, prepared angle diagrams

Physical materials needed: protractors, angle diagrams

Learning intention	Success criteria
Students are learning to: measure angles of up to 360° using a protractor	Students can: * I can line up the protractor's centre on the vertex and the baseline on one arm. * I can read the correct scale (inner or outer) to measure the angle. * I can measure reflex angles (measure the other angle and subtract from 360°).

Mini masterclass structure (30 min)

Activate (4 min): Show a protractor. Ask: where do you put the centre and the zero line?

Model (12 min): Place the centre on the vertex, baseline on one arm, read where the other arm crosses. Demonstrate choosing inner vs outer scale, and measuring a reflex angle via 360 – (the smaller angle).

Guided practice (10 min): Students measure a set of acute, obtuse and reflex angles with a protractor.

Check (4 min): Exit question: measure the given angle and say its type.

Key vocabulary	Differentiation
protractor, vertex, arm, baseline, scale, degrees, reflex	Support: Pre-mark the vertex and baseline; start with angles on the outer scale only. Extension: Measure reflex angles and check by measuring both angles to confirm they total 360°.
Assessment	Teaching notes (annotate here)
Students measure angles — links to R2 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 3 Drawing angles with a protractor [HANDS-ON]

R3 | Red band | DoE Lessons 6 & 7 (adapted)

Duration: 30 min | **Outcomes:** R3 | Syllabus: MA3-GM-03, MAO-WM-01 | GM-A: create angles up to 360° with a protractor

Linked to: DoE Lessons 6 & 7 (adapted) -- Lessons 6 & 7 -- Creating, identifying and investigating angles | Key resources: Protractors, rulers

Physical materials needed: protractors, rulers

Learning intention	Success criteria
Students are learning to: create (draw) angles of up to 360° using a protractor	Students can: * I can draw a baseline and mark the vertex. * I can use the protractor to mark the correct number of degrees and draw the second arm. * I can draw reflex angles by drawing the smaller angle and marking the reflex side.

Mini masterclass structure (30 min)

Activate (3 min): Ask students how they would draw an angle of exactly 60° .

Model (12 min): Draw a baseline, place the protractor centre on the vertex, mark 60° (or 110°), then join to make the second arm. Show drawing a reflex angle (e.g. 290°).

Guided practice (11 min): Students draw angles of given sizes (acute, obtuse and reflex) with a protractor.

Check (4 min): Exit question: draw a 135° angle and label its type.

Key vocabulary	Differentiation
protractor, draw, create, baseline, vertex, arm, degrees, reflex	Support: Provide the baseline and vertex already drawn so students focus on the protractor. Extension: Draw an angle, swap with a partner, and measure each other's angles to check.
Assessment	Teaching notes (annotate here)
Students draw angles — links to R3 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 4 Angles on a straight line and at a point

R4 | Red band | DoE Lessons 5, 6 & 8 (adapted)

Duration: 30 min **Outcomes:** R4 Syllabus: MA3-GM-03, MAO-WM-01 | GM-B: 90° / 180° / 360°; perpendicular

Linked to: DoE Lessons 5, 6 & 8 (adapted) -- Lessons 5, 6 & 8 -- Investigating angles | Key resources: Angle diagrams, straws or geostrips

Learning intention		Success criteria
Students are learning to: find missing angles using right angles, angles on a straight line and angles at a point		Students can: * I know perpendicular lines meet at 90° and angles on a straight line add to 180°. * I know angles at a point add to 360°. * I can find a missing angle by subtracting from 90°, 180° or 360°.
Mini masterclass structure (30 min)		
Activate (4 min): Show two angles on a straight line, one is 120°. Ask for the other. Model (12 min): Straight line: $180 - 120 = 60^\circ$. Right angle: $90 - \text{one part}$. Point: $360 - \text{the sum of the others}$. Show perpendicular lines making four 90° angles. Guided practice (10 min): Students find missing angles on straight lines, at right angles and at a point. Check (4 min): Exit question: three angles at a point are 120°, 150° and ?. Find the missing angle.		
Key vocabulary		Differentiation
straight line, point, right angle, perpendicular, adjacent, revolution, missing angle		Support: Provide the relevant total (180° or 360°) labelled on the diagram. Extension: Find two unknown equal angles that complete a straight line or point.
Assessment		Teaching notes (annotate here)
Students find missing angles — links to R4 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Mini Lesson 5 Identifying angles in diagrams

Y1 | Yellow band | DoE Lessons 5 & 8 (adapted)

Duration: 25-30 min **Outcomes:** Y1 Syllabus: MA3-GM-03, MAO-WM-01 | GM-B: identify angle types and parts in diagrams

Linked to: DoE Lessons 5 & 8 (adapted) -- Lessons 5 & 8 -- Identifying angles in diagrams | Key resources: Diagrams of intersecting lines

Learning intention

Students are learning to:
identify angle types and the parts of an angle in diagrams

Success criteria

Students can:

- * I can find right angles, angles on a straight line and angles at a point in a diagram.
- * I can name the vertex and the arms of an angle.
- * I can identify angle types formed when straight lines cross.

Mini masterclass structure (25-30 min)

Activate (4 min): Show two crossing lines. Ask students to point to a vertex and the arms of one angle.

Model (10 min): Label the vertex and arms; mark the right angles, straight-line angles (180°) and the angles at the point (360°) in the diagram.

Guided practice (10 min): Students label angle parts and identify angle types in diagrams of intersecting lines.

Check (4 min): Exit question: in a diagram of crossing lines, which angles sit on a straight line together?

Key vocabulary

vertex, arm, intersecting lines, right angle, straight line, point, angle type

Differentiation

Support: Provide a labelled example diagram to refer to.
Extension: Identify vertically opposite angles in the diagram and explain why they are equal.

Assessment

Students identify angles in diagrams — links to Y1 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 6 Transformations: translate, reflect, rotate [HANDS-ON]

Y2 | Yellow band | DoE Lessons 3 & 4 (adapted)

Duration: 30 min | **Outcomes:** Y2 | Syllabus: MA3-2DS-01, MAO-WM-01 | 2DS-B: translate/reflect/rotate; size preserved

Linked to: DoE Lessons 3 & 4 (adapted) -- Lessons 3 & 4 -- Translations, rotations and reflections | Key resources: Paper shapes, mirrors, grid paper

Physical materials needed: paper shapes, mirrors, grid paper

Learning intention	Success criteria
Students are learning to: use translate, reflect and rotate to describe transformations of 2D shapes	Students can: * I can slide a shape (translate), flip it (reflect) and turn it (rotate). * I can use the words translate, reflect and rotate to describe a move. * I know a transformation changes position and orientation but not the size of the shape.

Mini masterclass structure (30 min)

Activate (4 min): Slide, flip and turn a paper shape. Ask students to name each move.

Model (12 min): Translate (slide), reflect (flip over a line, like a mirror) and rotate (turn about a point). Show the shape stays the same size and shape each time.

Guided practice (10 min): Students perform and describe translations, reflections and rotations on grid paper.

Check (4 min): Exit question: which transformation flips a shape over a line?

Key vocabulary	Differentiation
translate, reflect, rotate, transformation, position, orientation, size	Support: Use a mirror for reflections and a pin to rotate about a point. Extension: Combine two transformations and describe the overall move.
Assessment	Teaching notes (annotate here)
Students describe transformations — links to Y2 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 7 Dissecting and rearranging shapes [HANDS-ON]

Y3 | Yellow band | DoE Lesson 3 (adapted)

Duration: 25-30 min

Outcomes: Y3

Syllabus: MA3-2DS-01, MAO-WM-01 | 2DS-B: dissect and rearrange one shape to make another

Linked to: DoE Lesson 3 (adapted) -- Lesson 3 -- Dissecting and rearranging shapes | Key resources: Paper shapes, scissors, tangram pieces

Physical materials needed: paper shapes, scissors, tangram pieces

Learning intention

Students are learning to:
dissect and rearrange one shape to make another

Success criteria

Students can:

- * I can cut a shape into pieces.
- * I can rearrange the pieces (using slides, flips and turns) to make a new shape.
- * I know the total area stays the same when I rearrange the pieces.

Mini masterclass structure (25-30 min)

Activate (4 min): Show a rectangle. Ask: how could we cut it and rearrange it into a different shape?

Model (10 min): Cut a rectangle along a diagonal and rearrange the two triangles into a different shape. Note the area is unchanged. Use tangram pieces to make a square then a different figure.

Guided practice (10 min): Students cut and rearrange shapes to make new shapes, using translations, reflections and rotations.

Check (4 min): Exit question: if you rearrange the pieces of a shape, does its area change?

Key vocabulary

dissect, rearrange, pieces, transform, area, translate, reflect, rotate

Differentiation

Support: Provide pre-cut pieces (e.g. a tangram) to rearrange.
Extension: Dissect a parallelogram and rearrange it into a rectangle; explain what stayed the same.

Assessment

Students dissect and rearrange shapes — links to Y3 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 8 Calculating elapsed time

Y4 | Yellow band | DoE Lesson 1 (adapted)

Duration: 30 min | **Outcomes:** Y4 | Syllabus: MA3-NSM-02, MAO-WM-01 | NSM-B: elapsed time; bridging; rounding

Linked to: DoE Lesson 1 (adapted) -- Lesson 1 -- Rounding time calculations | Key resources: Clock faces, number lines

Learning intention		Success criteria
Students are learning to: calculate elapsed time using start and finish times and bridging		Students can: * I can use start and finish times to work out how long something lasts. * I can add and subtract time by bridging to the next hour. * I can round a time answer to the nearest minute or hour.
Mini masterclass structure (30 min)		
Activate (4 min): A film starts at 2:40 pm and ends at 4:15 pm. How long is it? Model (12 min): Bridge: 2:40 → 3:00 is 20 min; 3:00 → 4:15 is 1 h 15 min; total 1 h 35 min. Show rounding 1 h 33 min to the nearest hour ($\approx$ 1 h 30 min or 2 h). Guided practice (10 min): Students calculate elapsed time using bridging on a number line and round their answers. Check (4 min): Exit question: a bus leaves at 9:50 am and arrives at 11:20 am. How long is the trip?		
Key vocabulary		Differentiation
elapsed time, duration, start, finish, bridging, round, hour, minute		Support: Use a number line of times and bridge to the next hour first. Extension: Calculate elapsed time that crosses midday or midnight.
Assessment		Teaching notes (annotate here)
Students calculate elapsed time — links to Y4 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:

Mini Lesson 9 Time intervals as decimals

Y5 | Yellow band | DoE Lesson 2 (adapted)

Duration: 25 min **Outcomes:** Y5 Syllabus: MA3-NSM-02, MAO-WM-01 | NSM-B: represent time intervals as decimals

Linked to: DoE Lesson 2 (adapted) -- Lesson 2 -- Time intervals as decimals | Key resources: Clock faces, fraction-of-an-hour cards

Learning intention		Success criteria	
Students are learning to: represent commonly used time intervals as decimals		Students can: * I know 30 minutes is half an hour = 0.5 h. * I know 15 minutes = 0.25 h and 45 minutes = 0.75 h. * I can write a number of minutes as a decimal of an hour (minutes ÷ 60).	
Mini masterclass structure (25 min)			
Activate (3 min): Ask: half an hour is how many minutes? What decimal of an hour is that?			
Model (10 min): 30 min = 30/60 = 0.5 h; 15 min = 0.25 h; 45 min = 0.75 h; 6 min = 0.1 h. Connect to benchmark fractions.			
Guided practice (8 min): Students write common time intervals as decimals of an hour.			
Check (4 min): Exit question: write 45 minutes as a decimal of an hour.			
Key vocabulary		Differentiation	
interval, decimal, hour, minute, half, quarter, fraction of an hour		Support: Provide a table linking 15/30/45 min to 0.25/0.5/0.75 h. Extension: Convert decimals back to minutes (e.g. 0.2 h = 12 min).	
Assessment		Teaching notes (annotate here)	
Students represent time as decimals — links to Y5 in the SOLO Show task.		Taught as described / adapted / replaced -- note here:	

Mini Lesson 10 Duration in 12- and 24-hour time

Y6 | Yellow band | DoE Lesson 2 (adapted)

Duration: 30 min | **Outcomes:** Y6 | Syllabus: MA3-NSM-02, MAO-WM-01 | NSM-B: 12-/24-hour duration problems

Linked to: DoE Lesson 2 (adapted) -- Lesson 2 -- Time intervals as decimals (12/24-hour problems) | Key resources: 24-hour time charts, timetables

Learning intention

Students are learning to:
convert between 12- and 24-hour time and solve duration problems

Success criteria

Students can:

- * I can convert a 12-hour time (with am/pm) to 24-hour time and back.
- * I can read a timetable in 24-hour time.
- * I can solve a duration problem using 24-hour time.

Mini masterclass structure (30 min)

Activate (4 min): What is 3:30 pm in 24-hour time? (15:30) How did you work it out?

Model (12 min): After 12 noon, add 12 to the hour for 24-hour time (3 pm = 15:00). Read a bus timetable and find the duration between two 24-hour times.

Guided practice (10 min): Students convert times and solve duration problems using 24-hour notation.

Check (4 min): Exit question: a train leaves at 13:45 and arrives at 15:20. How long is the journey?

Key vocabulary

12-hour, 24-hour, am, pm, timetable, duration, convert

Differentiation

Support: Provide a 12-to-24-hour conversion chart.
Extension: Plan a short journey using a real 24-hour timetable and calculate total travel time.

Assessment

Students solve 24-hour duration problems — links to Y6 in the SOLO Show task.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 11 Complementary, supplementary and vertically opposite angles

G1 | Green band | Original mini lesson (Stage 4 extension)

Duration: 30 min **Outcomes:** G1 Syllabus: MA4-ANG-C-01, MAO-WM-01 | Stage 4: complementary / supplementary / vertically opposite

Learning intention

Students are learning to:
use complementary, supplementary and vertically opposite angles to find unknown angles

Success criteria

Students can:

- * I know complementary angles add to 90° and supplementary angles add to 180° .
- * I know vertically opposite angles are equal.
- * I can find an unknown angle using these relationships.

Mini masterclass structure (30 min)

Activate (4 min): Two angles add to 90° . If one is 35° , what is the other?

Model (12 min): Complementary: $90 - 35 = 55^\circ$. Supplementary: $180 -$ one angle. Show two crossing lines: vertically opposite angles are equal.

Guided practice (10 min): Students find unknown angles using complementary, supplementary and vertically opposite relationships.

Check (4 min): Exit question: the supplement of 110° is what angle?

Key vocabulary

complementary, supplementary, vertically opposite, equal, unknown angle

Differentiation

Support: Provide the relationship ($90^\circ/180^\circ$) labelled on each diagram.
Extension: Find several unknown angles in a single diagram of intersecting lines.

Assessment

Students apply angle relationships — extends R4/Y1.

Teaching notes (annotate here)

Taught as described / adapted / replaced -- note here:

Mini Lesson 12 Angles on parallel lines cut by a transversal

G2 | Green band | Original mini lesson (Stage 4 extension)

Duration: 30 min | **Outcomes:** G2 | Syllabus: MA4-ANG-C-01, MAO-WM-01 | Stage 4: corresponding / alternate / co-interior

Learning intention		Success criteria	
Students are learning to: identify corresponding, alternate and co-interior angles on parallel lines		Students can: * I can identify a transversal crossing two parallel lines. * I know corresponding and alternate angles are equal, and co-interior angles add to 180°. * I can use these to find unknown angles.	
Mini masterclass structure (30 min)			
Activate (4 min): Show two parallel lines crossed by a transversal. Point out matching (corresponding) angles. Model (12 min): Corresponding angles equal; alternate (Z) angles equal; co-interior (C) angles add to 180°. Use them to find unknown angles. Guided practice (10 min): Students identify and use corresponding, alternate and co-interior angles to find unknowns. Check (4 min): Exit question: a co-interior angle is 70°. What is the other co-interior angle?			
Key vocabulary		Differentiation	
parallel, transversal, corresponding, alternate, co-interior, equal, supplementary		Support: Colour-code each angle pair on the diagram. Extension: Find every angle in a parallel-line diagram given just one angle.	
Assessment		Teaching notes (annotate here)	
Students use parallel-line angle relationships — extends Y1.		Taught as described / adapted / replaced -- note here:	

Mini Lesson 13 Solving numerical angle problems G3 Green band Original mini lesson (Stage 4 extension)		
Duration: 30 min	Outcomes: G3	Syllabus: MA4-ANG-C-01, MAO-WM-01 Stage 4: solve numerical angle problems with reasoning
Learning intention		Success criteria
Students are learning to: solve numerical angle problems by combining angle relationships and giving reasons		Students can: * I can choose the right angle relationship to use. * I can combine relationships to find an unknown angle in steps. * I can give a reason for each step.
Mini masterclass structure (30 min)		
Activate (4 min): A diagram has an unknown angle that needs two steps. What relationships could help? Model (12 min): Work a multi-step problem: use angles on a straight line (180°) to find one angle, then vertically opposite to find another, giving a reason each time. Guided practice (10 min): Students solve multi-step angle problems, writing a reason for each step. Check (4 min): Exit question: find x using two angle relationships and state your reasons.		
Key vocabulary		Differentiation
reasoning, relationship, straight line, point, vertically opposite, supplementary, reason		Support: Prompt with 'which angles add to 180° or 360° here?' for each step. Extension: Create a multi-step angle puzzle for a partner and provide a worked solution.
Assessment		Teaching notes (annotate here)
Students solve numerical angle problems — extends R4/G1.		Taught as described / adapted / replaced -- note here:

Unit 39 -- Resource Appendix

Every link verified. Videos open in a browser; worksheets are print-ready PDFs (see the download checklist at the end).

Type	Resource	Link
R1 -- Classify angles as acute, right, obtuse, straight or reflex.		
Video	Types of Angles — Acute / Obtuse / Right / Straight / Reflex Math with Mr. J	https://www.youtube.com/watch?v=UsE1hu-q0Cs
Video	How to Use a Protractor to Measure an Angle Math with Mr. J	https://www.youtube.com/watch?v=N8TGOMH5Y50
Worksheet (print)	Corbettmaths — Types of Angles (PDF)	https://corbettmaths.com/wp-content/uploads/2018/09/Types-of-Angle-pdf.pdf
R2 -- Measure angles of up to 360° using a protractor.		
Video	How to Measure Angles Using a Protractor Math with Mr. J	https://www.youtube.com/watch?v=j5D7acsExxw
Video	How to Use a Protractor — Measuring and Drawing Angles Math with Mr. J	https://www.youtube.com/watch?v=2on99N5jBJ4
Worksheet (print)	Corbettmaths — Measuring Angles (PDF)	https://corbettmaths.com/wp-content/uploads/2018/02/measuring-angles.pdf
R3 -- Create (draw) angles of up to 360° using a protractor.		
Video	How to Draw Angles with a Protractor Math with Mr. J	https://www.youtube.com/watch?v=zh-tj7oTQlc
Video	Drawing an Angle with a Protractor — 60° Math with Mr. J	https://www.youtube.com/watch?v=qYfH4NN0b-4
Worksheet (print)	Corbettmaths — Constructing / Drawing Angles (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/constructing-angles-pdf.pdf
R4 -- Find missing angles on a straight line, at a point and at a right angle.		
Video	Angles at a Point Corbettmaths	https://www.youtube.com/watch?v=mdAwUsf0k1s
Video	Angles in a Straight Line Corbettmaths	https://www.youtube.com/watch?v=q5tV5V56Hr0
Worksheet (print)	Corbettmaths — Angle Facts (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/angle-facts-pdf1.pdf
Y1 -- Identify angle types and parts (vertex/arms) in diagrams.		
Video	Angles on a Straight Line and About a Point Mr Mathematics	https://www.youtube.com/watch?v=juix_jB1pss
Video	Finding the Missing Angle of a Triangle Math with Mr. J	https://www.youtube.com/watch?v=LAIQnW6-6c4
Worksheet (print)	Corbettmaths — Missing Angles (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/missing-angles-pdf1.pdf
Y2 -- Describe transformations (translate, reflect, rotate).		
Video	Transformations — Translations / Reflections / Rotations Math Defined	https://www.youtube.com/watch?v=YD3HIMUae_4
Video	3 Types of Transformations — Translations Reflections & Rotations Turtlediary	https://www.youtube.com/watch?v=VJTxxv-tRKj0
Worksheet (print)	Corbettmaths — Reflections (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/reflections-pdf.pdf
Y3 -- Dissect and rearrange one shape to make another.		
Video	Compose and Decompose 2D Shapes Underwater Math	https://www.youtube.com/watch?v=KbDXgDeFoG4
Video	3 Types of Transformations — Translations Reflections & Rotations Turtlediary	https://www.youtube.com/watch?v=VJTxxv-tRKj0
Worksheet (print)	Corbettmaths — Rotations (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/rotations-pdf.pdf
Y4 -- Calculate elapsed time using bridging and rounding.		
Video	Elapsed Time — T-Chart Strategy Sally Osborne	https://www.youtube.com/watch?v=4Eucvb7qFpk
Worksheet (print)	Corbettmaths — Time Calculations (PDF)	https://corbettmaths.com/wp-content/uploads/2019/10/Time-Calculations-pdf.pdf
Worksheet (print)	Corbettmaths Primary — Time (PDF)	https://corbettmathsprimary.com/wp-content/uploads/2018/07/time-pdf.pdf
Y5 -- Represent commonly used time intervals as decimals.		

Type	Resource	Link
Video	Converting Fractions and Decimals to Hours and Minutes FountainMath	https://www.youtube.com/watch?v=r6Z48lv1HC
Video	How To Convert Decimal Time To Hours And Minutes OnTheClock	https://www.youtube.com/watch?v=IPaPdf9ck8M
Worksheet (print)	Corbettmaths — Time Calculations (PDF)	https://corbettmaths.com/wp-content/uploads/2019/10/Time-Calculations-pdf.pdf
Y6 -- Solve duration problems in 12- and 24-hour time.		
Video	The 24-Hour Clock Mathematics Grade 5 Periwinkle	https://www.youtube.com/watch?v=Lj4YxVZVLVU
Video	Understanding the 24-Hour Clock Smartli	https://www.youtube.com/watch?v=dQ5vIHA4L5s
Worksheet (print)	Corbettmaths — Timetables and Time (PDF)	https://corbettmaths.com/wp-content/uploads/2025/03/Timetables-and-Time-Practice-Questions.pdf
G1 -- Complementary, supplementary and vertically opposite angles.		
Video	Complementary Angles & Supplementary Angles Math with Mr. J	https://www.youtube.com/watch?v=Qpq_XEeBBZw
Video	Complementary and Supplementary Angles — How to Find Missing Angles Math with Mr. J	https://www.youtube.com/watch?v=E_uIN2FjJac
Worksheet (print)	Corbettmaths — Angle Facts (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/angle-facts-pdf1.pdf
G2 -- Angles on parallel lines cut by a transversal.		
Video	Angles in Parallel Lines Corbettmaths	https://www.youtube.com/watch?v=5NfXvBdwKJE
Video	Parallel Lines — Angle Examples Corbettmaths	https://www.youtube.com/watch?v=ARsZDZIPdHM
Worksheet (print)	Corbettmaths — Angles in Parallel Lines (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/angles-in-parallel-lines-pdf1.pdf
G3 -- Solve numerical angle problems using reasoning.		
Video	What are Complementary Angles? — Find Missing Angles Math with Mr. J	https://www.youtube.com/watch?v=odR4R_BPiH0
Video	What are Supplementary Angles? — Find Missing Angles Math with Mr. J	https://www.youtube.com/watch?v=rkGQ7O2i4-U
Worksheet (print)	Corbettmaths — Missing Angles (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/missing-angles-pdf1.pdf
Beyond SOLO -- Stage 4 extension		
Video	Complementary Angles & Supplementary Angles Math with Mr. J	https://www.youtube.com/watch?v=Qpq_XEeBBZw
Video	Angles in Parallel Lines Corbettmaths	https://www.youtube.com/watch?v=5NfXvBdwKJE
Worksheet (print)	Corbettmaths — Angles in Parallel Lines (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/angles-in-parallel-lines-pdf1.pdf
Worksheet (print)	Corbettmaths — Missing Angles (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/missing-angles-pdf1.pdf
Worksheet (print)	Corbettmaths — Angle Facts (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/angle-facts-pdf1.pdf

Worksheet download checklist

Download and print these PDFs before the unit (or save to your class drive). Tick each as you go.

- ☐ Corbettmaths — Types of Angles (PDF) -- <https://corbettmaths.com/wp-content/uploads/2018/09/Types-of-Angle-pdf.pdf>
- ☐ Corbettmaths — Measuring Angles (PDF) -- <https://corbettmaths.com/wp-content/uploads/2018/02/measuring-angles.pdf>
- ☐ Corbettmaths — Constructing / Drawing Angles (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/constructing-angles-pdf.pdf>
- ☐ Corbettmaths — Angle Facts (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/angle-facts-pdf1.pdf>
- ☐ Corbettmaths — Missing Angles (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/missing-angles-pdf1.pdf>
- ☐ Corbettmaths — Reflections (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/reflections-pdf.pdf>
- ☐ Corbettmaths — Rotations (PDF) -- <https://corbettmaths.com/wp-content/uploads/2013/02/rotations-pdf.pdf>
- ☐ Corbettmaths — Time Calculations (PDF) -- <https://corbettmaths.com/wp-content/uploads/2019/10/Time-Calculations-pdf.pdf>
- ☐ Corbettmaths Primary — Time (PDF) -- <https://corbettmathsprimary.com/wp-content/uploads/2018/07/time-pdf.pdf>
- ☐ Corbettmaths — Timetables and Time (PDF) -- <https://corbettmaths.com/wp-content/uploads/2025/03/Timetables-and-Time-Practice-Questions.pdf>

